

*MA.8.AR.4.2***Benchmark**

MA.8.AR.4.2 **Given a system of two linear equations represented graphically on the same coordinate plane, determine whether there is one solution, no solution or infinitely many solutions.**

Connecting Benchmarks/Horizontal Alignment

- MA.8.AR.2.1
- MA.8.AR.3.4, MA.8.AR.3.5
- MA.8.AR.4.1, MA.8.AR.4.3

Terms from the K-12 Glossary

- Linear Equation

Vertical Alignment**Previous Benchmarks**

- MA.7.AR.4.3, MA.7.AR.4.4, MA.7.AR.4.5

Next Benchmarks

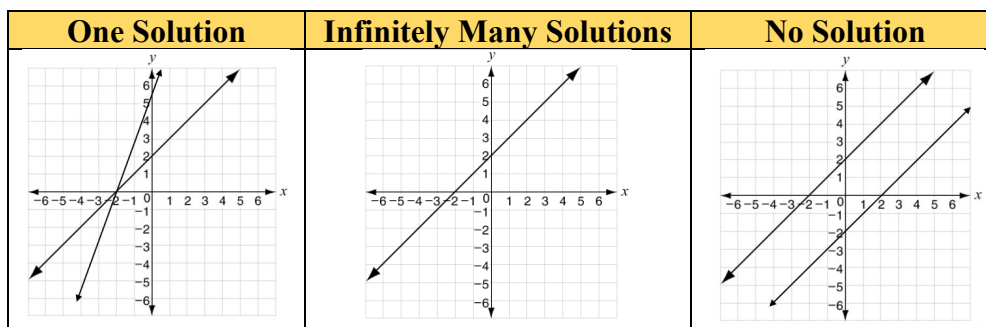
- MA.912.AR.9.1, MA.912.AR.9.4

Purpose and Instructional Strategies

In grade 7, students determined constants of proportionality and graphed proportional relationships from a table, equation or a written description given in a mathematical or real-world context. In grade 8, students extend this learning to a system of two linear equations represented graphically on the same coordinate plane then students will determine whether there is one solution, no solution or infinitely many solutions. In high school, students will write and solve a system of two-variable linear equations algebraically and graphically given a mathematical or real-world context.

- The focus of this benchmark is on the understanding that the solution of a system is a set of points that satisfy both equations of the system.
- Systems of linear equations can have one solution, infinitely many solutions or no solutions.
 - A system of linear equations whose graphs meet at one point (intersecting lines) has only one solution, the ordered pair representing the point of intersection.
 - A system of linear equations whose graphs are coincident (the same line) has infinitely many solutions, the set of ordered pairs representing all the points on the line.

- A system of linear equations whose graphs do not meet (parallel lines) has no solutions and the slopes of these lines are the same. The technical name for these kinds of systems is "inconsistent."



- A system of linear equations is two linear equations that should be solved at the same time.
- Instruction includes understanding that systems are on the same coordinate plane to determine solutions (*MTR.4.1*).

Common Misconceptions or Errors

- Students may incorrectly interpret the solution when the lines are the same and have an infinite number of solutions. To address this misconception, provide multiple examples to show how the equations and graphs will be the same line on the coordinate plane.

Strategies to Support Tiered Instruction

- Instruction includes testing possible solutions for a given system of linear equations to demonstrate whether the equations have the same solution set, one common solution (only one ordered pair) or no common solution.
- Instruction includes drawing connections between systems of equations represented graphically and with equations. Using a graphic organizer, reinforce the solution to a system of equations as the ordered pair that stratifies both equations simultaneously.
 - When there is one solution, the two lines intersect at one point and when using substitution, the coordinates of that one point will result in true statements for both equations.
 - When there is no solution, the two lines do not intersect and therefore there are no coordinates that will result in true statements for both equations.
 - When there are infinite solutions, the two lines coincide and intersect with an infinite number of points. When substituting ordered pairs, all will result in true statements for both equations.

Instructional Tasks

Instructional Task 1 (MTR.6.1)

Ashley looks at the following system of equations. She concludes that because there is no y -intercept value, the lines cannot intersect.

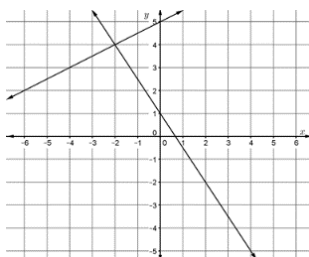
$$y = -\frac{1}{2}x \quad y = \frac{1}{3}x$$

Part A. Graph the system of equations on a coordinate plane.

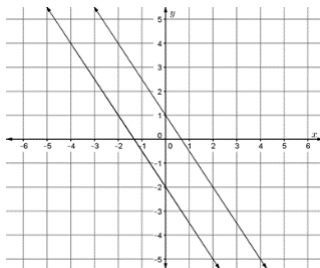
Part B. Is Ashley's conclusion correct? Explain your answer and support your reasoning with mathematical examples.

Instructional Task 2 (MTR.3.1)

Part A. Identify the solution of the graphed system of equations. Explain why it is the solution.



Part B. Identify the solution of the graphed system of equations. Explain how you know it is the solution. What conjecture can you make about parallel lines?

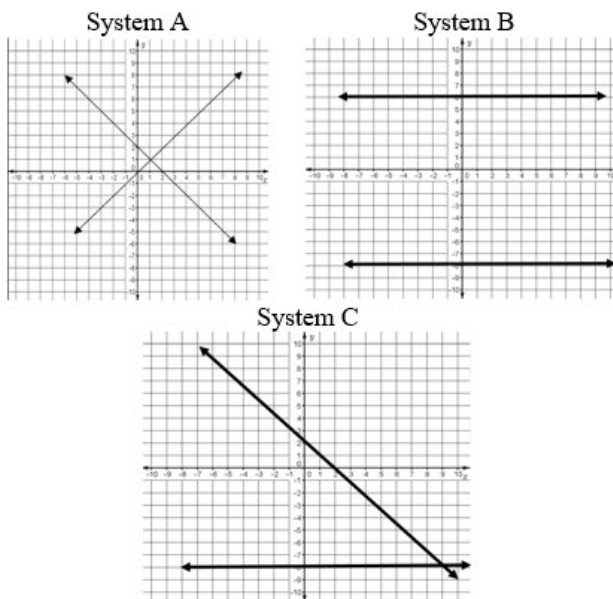


Instructional Items

Instructional Item 1

Determine the number of solutions of each graphed system of linear equations, A, B and C.

System A	System B	System C



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*